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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	MLP
Selected Domain	Domain 1 - AIVERSECITY.
Selected SDG Track (1–17)	4,5,8,9,10
Project / Solution Name	Skillcast
Name (Leader)	Omirzak Aiym
Email & Phone (Leader)	umerzakova.aiym@gmail.com/ +60178089335
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing workplace tasks and the skills employers need. However, graduates, working adults and career changers often struggle to translate these changes into a practical decision: Which career direction suits me, and what should I learn next?

The World Economic Forum's Future of Jobs Report 2025 estimates that 39% of workers' existing skills will change or become outdated between 2025 and 2030, while 59 out of every 100 workers will require training. These are global employer-survey benchmarks, rather than measurements of Malaysian workers. SkillCast focuses on this challenge in Malaysia. Career planning requires users to connect information scattered across job boards, labour-market reports and course websites. A vacancy describes an employer's requirements, while a course catalogue lists learning options; users must still determine which opportunities fit their experience, identify missing skills and choose relevant training within their available time.

Employers and career advisers face a related challenge: identifying training priorities and turning skills assessments into actionable learning plans. People with accessibility needs, limited budgets or little time for study may face additional barriers.

The social concern is unequal access to useful career guidance and learning opportunities. The economic concern is misdirected training effort and persistent skills mismatches. Sustainability also requires people equipped for roles supporting areas such as clean energy and responsible production. The problem is therefore not simply access to information, but the difficulty of turning changing evidence into an achievable next step.

Why it matters:

1. Social: without guidance, workers risk displacement and growing inequality (SDG 8, 10, 5).
2. Economic: skills mismatch limits productivity and innovation (SDG 8, 9).
3. Environmental: many future jobs in clean energy, health, smart agriculture and responsible AI will help deliver the UN SDGs, but they cannot be filled without the right skills (SDG 4, 7, 13).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SkillCast is an AI-assisted career guidance platform for graduates, working adults and career changers navigating a changing job market. It connects a user's existing skills, interests and available learning time with career options and a practical plan for developing missing skills.

Users begin with a questionnaire or, for demonstration, a synthetic CV. Qwen2.5-7B-Instruct, hosted through vLLM on RunPod, extracts structured information from CV text after browser-side removal of identifiable details. Users review and correct the extracted profile before confirming it. The model also turns calculated career-match results into plain-language explanations.

The platform combines this AI assistance with transparent, rule-based calculations. It compares the confirmed profile against curated role requirements to identify career matches and skill gaps. Public job-posting signals and cited research inform skill-demand indicators, while two AI-adoption scenarios present alternative outlooks rather than guaranteed predictions. Fallback extraction and explanation templates keep the core journey available when the model cannot respond.

SkillCast's distinguishing feature is the connection between career exploration and achievable action. Once a user selects a role, a personalized Harada roadmap organises learning into eight areas with practical tasks, relevant courses from a curated catalogue, and estimated study time based on weekly availability. Employers can also explore team skill gaps using synthetic workforce profiles.

Applying technical skills covered in SDC Phase 1, the team integrates GenAI APIs, structured JSON outputs, database design and cloud inference. The prototype uses Next.js, React and TypeScript, with Prisma and SQLite storing market-signal snapshots.

The intended outcome is clearer career decisions and less time spent connecting scattered job-market information with suitable learning resources.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend (Next.js App Router, React, TypeScript)

Interface and accessibility (Tailwind CSS, shadcn/ui, next-themes)

Backend (Next.js API routes)-handle CV extraction, AI explanations and labour-market signal ingestion

Generative AI (Qwen2.5 - 7B-Instruct served through vLLM on RunPod) - extract structured information from de-identified CV text, subject to user confirmation.Used RunPod mcp connected to DeepSeek V4 Flash with OMP harness API for which we got from OpenRouter, to have it set up vLLM with Qwen2.5-7B-Instruct running on a 24 GB GPU

Matching/Planning (TypeScript rule-based engines) - calculate explainable compatibility scores, skill gaps, momentum indicators and learning estimates.

Database (Prisma ORM and SQLite) - store signal runs, skill snapshots and update history.

Deployment (Apache Tomcat, Next.js , Bun standalone output and Caddy) - run the application on a cloud server and route incoming web requests.

Collaboration (GitHub branches and pull requests, Replit) - coordinate development, review changes and integrate team contributions.

Labour-market data (Remotive,Arbeitnow,Greenhouse and Ashby APIs;curated research) - supply job posting signals and research context, with cached or curated fallbacks.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied the Design Thinking methodology through five stages: Empathize, Define, Ideate, Prototype, and Test.

Empathize & Define: We identified challenges faced by students, job seekers, professionals, and employers adapting to AI-driven workforce transformation, including job displacement risks, skill gaps, career uncertainty, and limited access to personalized reskilling guidance.

Ideate: We designed SkillCast as a human-centered AI platform integrating a Future Skills Radar, explainable career matching, personalized learning pathways, and the Harada Method to turn career recommendations into achievable goals.

Prototype: We developed an interactive web application featuring CV-based skill profiling, career match scores, skill-gap analysis, progress tracking, and SDG alignment. Our modular architecture supports integration with continuously updated AI trends and labor-market data.

Test & Deployment: We prioritized intuitive onboarding, responsive design, inclusivity, multilingual accessibility, and transparent recommendations. In Phase 2, we plan to conduct usability testing, integrate verified and regularly updated data sources, evaluate matching accuracy, and strengthen privacy safeguards. To address prediction uncertainty, we will introduce multiple AI-disruption scenarios and clearly communicate their limitations.

By combining user-centered design, technical feasibility, and responsible AI, SkillCast aims to become a scalable, reliable platform empowering individuals to make informed career decisions.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG Target 4.4 - Skills for employment. SkillCast supports this target by identifying skill gaps and connecting users with relevant learning resources and practical Harada roadmap tasks. Its immediate contribution is helping users prepare a learning plan; actual skill development requires subsequent learning and assessment. Secondary: SDG Target 8.5 -Productive employment and decent work for all. SkillCast supports career preparation for graduates, working adults and career changers. Employment outcomes and benefits for users with disabilities will require further evaluation.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Our provisional baseline is 60 minutes per person to manually identify three suitable career options, review relevant skill-demand evidence, and prepare a learning plan for one selected role, including skill gaps, relevant course links and estimated study time. This process requires comparing information across job boards, research sources and course catalogues. The figure is an unvalidated team estimate, rather than a published benchmark or measured average. Phase 2 will establish a measured baseline using timed manual tasks with synthetic profiles.
Core Measurable KPI *The specific unit you will track*	Average time, measured in minutes, from starting profile entry to obtaining three ranked career recommendations and reviewing a personalised learning plan for one selected role,

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	including identified skill gaps, relevant course links and estimated study time.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve at least an 80% reduction in average career-and-learning planning time compared with the measured manual baseline. Against the provisional 60-minute estimate, this corresponds to 12 minutes or less. One internal end-to-end prototype test recorded 4 minutes 36 seconds. This preliminary result demonstrates feasibility but does not establish an average or a verified percentage reduction, as the manual baseline remains unvalidated. During Phase 2, at least six participants will compare manual planning and SkillCast using equivalent synthetic profiles and identical completion criteria. The order of methods will be counterbalanced to reduce familiarity effects. Timing will include profile entry, processing and review of recommendations and the learning plan. A run will count as complete when career recommendations, skill gaps, relevant course links and estimated study time are available, with course relevance checked against the identified gaps. Incomplete runs will be reported separately. This evaluation measures planning efficiency and course discovery, rather than course completion, skill acquisition, employment gains or environmental savings.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

√ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

√ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 16.1.3	Code - web framework	MIT
React / React DOM 19.2.3	Code - UI libraries	MIT
Radix UI React packages (@radix-ui/react-*, listed in package.json)	Code - UI components	MIT
TanStack React Table 8.21.3	Code - tables	MIT
Prisma / Prisma Client 6.19.2	Code - database tooling	Apache-2.0, Bun

Class Variance Authority 0.7.1	Code - styling utility	Apache-2.0, Bun
Lucide React 0.525.0	Code -icons	ISC
Zustand 5.0.10	Code - state management	MIT
Zod 4.3.5	Code - schema validation	MIT
Recharts 2.15.4	Code - charts	MIT
React Hook Form 7.71.1	Code - forms	MIT
React Day Picker 9.13.0	Code - date picker	MIT
React Resizable Panels 3.0.6	Code - resizable layouts	MIT
next-themes 0.4.6	Code - theme switching	MIT
clsx 2.1.1; tailwind-merge 3.4.0; tailwindcss-animate 1.0.7	Code - styling utilities	MIT
cmdk 1.1.1; embla-carousel-react 8.6.0; input-otp 1.4.2; sonner 2.0.7; vaul 1.1.2	Code - interface components	MIT
Qwen2.5-7B-Instruct	Model - CV extraction and recommendation explanations	Apache-2.0, Bun
vLLM	Code - model inference server	Apache-2.0, Bun
SQLite	Code - database engine	Public domain
Remotive, Arbeitnow, Greenhouse and Ashby job postings	Dataset/API - labour-market signals	Respective provider/content-owner terms; no blanket open-data licence claimed
WEF Future of Jobs Report 2025; Hays Asia Salary Guide 2025; Stanford AI Index 2025; MOSTI estimate cited in the competition brief	Research/reference data	Respective publisher terms and attribution requirements
United Nations SDG targets	Reference data	UN terms of use and attribution requirements

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: 

Date: 24.10.2026